ClawFi: A Hedera■Native AI Treasury Workforce

Technical Whitepaper & System Documentation

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Abstract

ClawFi introduces a new paradigm for autonomous financial systems: AI-native treasury workforces. Instead of a single assistant interacting with a wallet, ClawFi deploys a coordinated network of specialized AI agents responsible for market research, DeFi strategy analysis, risk validation, execution planning, and reporting. These agents collaborate to form a transparent, auditable decision pipeline for capital allocation.

ClawFi integrates three core technologies: OpenClaw as the agent orchestration runtime, Hedera as the economic coordination and receipt layer, and a modular strategy engine for token trading and DeFi yield discovery. Together they enable autonomous financial intelligence that remains transparent, accountable, and human-controlled.

1. Introduction

Artificial intelligence has reached a stage where it can analyze markets, evaluate strategies, and assist with financial decisions. However, current AI-powered financial tools remain limited. Most systems simply attach a chatbot to a wallet or analytics dashboard.

ClawFi reimagines this architecture. Instead of relying on a single model, ClawFi deploys multiple specialized agents operating under a unified treasury policy. Each agent performs a distinct responsibility such as research, strategy evaluation, risk enforcement, or execution planning. By distributing tasks across specialized agents, ClawFi achieves higher transparency and safer decision-making.

Hedera provides the infrastructure layer that makes this model economically viable. Through consensus topics, scheduled transactions, token services, and mirror nodes, ClawFi can log agent decisions, coordinate actions, and settle micropayments while preserving full auditability.

2. Vision

ClawFi envisions financial systems where AI agents operate as a workforce rather than a single assistant. Each agent specializes in a narrow domain and collaborates through verifiable workflows.

This model mirrors real-world organizations where different roles contribute to financial decision-making.

In ClawFi, the coordinator agent orchestrates tasks. Research agents identify opportunities, strategy agents analyze potential yield sources, risk agents validate policy compliance, and execution agents prepare transaction flows. Human users remain in the loop through approval-based execution.

3. System Architecture

ClawFi consists of four primary layers: user interface, agent orchestration, strategy engines, and Hedera infrastructure. The user interface exposes treasury configuration, opportunity discovery, and strategy results. The orchestration layer manages agent workflows using the OpenClaw runtime. Strategy engines analyze market and protocol data. The Hedera layer records receipts, coordinates transactions, and enables micropayments.

4. Agent Architecture

Coordinator Agent: interprets user goals and decomposes them into subtasks for other agents.

Token Research Agent: scans tokens, evaluates liquidity, narratives, and risk signals to produce trade theses.

DeFi Strategy Agent: evaluates lending markets, vaults, staking opportunities, and liquidity pools to identify optimal yield allocations.

Risk Agent: enforces treasury policy such as exposure limits, liquidity requirements, and protocol risk thresholds.

Execution Agent: prepares transaction previews and simulation flows for user approval.

Reporter Agent: synthesizes outputs from all agents and produces clear reports for users.

5. Hedera Integration

ClawFi leverages Hedera’s infrastructure to provide verifiable coordination and transparent economic flows.

Hedera Consensus Service (HCS) acts as the coordination bus. Agent events such as task creation, assignment, completion, and allocation decisions are recorded in ordered topics.

Scheduled Transactions enable approval-based execution. AI prepares transactions, but users approve them before final execution.

Hedera Token Service (HTS) supports treasury tokens and agent reward distribution through custom fee mechanisms.

Mirror Nodes provide read-only access to historical receipts, enabling a powerful audit dashboard.

6. Treasury Model

ClawFi organizes capital into several pools: reserve capital, strategy capital, and agent reward budgets. Treasury policy controls how capital is distributed between token strategies and DeFi yield opportunities.

7. Strategy Engine

The token strategy engine identifies trade opportunities based on liquidity, narrative trends, and risk metrics. The DeFi strategy engine identifies yield opportunities across lending markets, staking protocols, and liquidity pools. Both engines feed into a unified allocation engine that produces a portfolio recommendation aligned with user-defined policy.

8. Agent Economy

ClawFi treats AI agents as economic participants. When agent work is accepted, small rewards can be distributed using Hedera payments. This creates a transparent economic model for autonomous work.

9. Security Model

Security is enforced through multiple layers: risk policies, approval-based execution, transparent receipts, and audit logs. This ensures that automation never removes human oversight.

10. Example Workflow

A user asks ClawFi to allocate idle capital. The coordinator agent decomposes the request. Research and strategy agents analyze opportunities. The risk agent validates constraints. The coordinator synthesizes a final plan. The execution agent prepares scheduled transactions. Once the user approves, execution occurs and receipts are recorded on Hedera.

11. Roadmap

Phase 1: Multi-agent orchestration and treasury analysis.

Phase 2: Advanced strategy engines and automated monitoring.

Phase 3: Open agent marketplace and third-party strategy agents.

12. Conclusion

ClawFi represents a new model for financial intelligence. By combining agent-based reasoning with Hedera’s verifiable infrastructure, it enables transparent, coordinated, and economically aligned AI decision-making. The result is an autonomous treasury workforce capable of managing financial strategies while remaining auditable and secure.